

A Nice Result for

Fibonacci Matrices

A nice result for Fibonacci matrices

$$f_0 = 0, f_1 = 1, f_{n+2} = f_{n+1} + f_n$$

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, \dots$$

observe that: $\det \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix} = -1$, $\det \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} = 1$

$$\det \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} = -1, \det \begin{pmatrix} 2 & 3 \\ 5 & 8 \end{pmatrix} = 1$$

$$\det \begin{pmatrix} 3 & 5 \\ 8 & 13 \end{pmatrix} = -1, \det \begin{pmatrix} 5 & 8 \\ 13 & 21 \end{pmatrix} = 1$$

claim For all $n \geq 0$, $\det \begin{pmatrix} f_n & f_{n+1} \\ f_{n+2} & f_{n+3} \end{pmatrix} = (-1)^{n+1}$

proof: by induction: Suppose for some $k \geq 0$, we have:

$$\det \begin{pmatrix} f_k & f_{k+1} \\ f_{k+2} & f_{k+3} \end{pmatrix} = (-1)^{k+1}.$$

Consider

$$\det \begin{pmatrix} f_{k+1} & f_{k+2} \\ f_{k+3} & f_{k+4} \end{pmatrix} = \det \begin{pmatrix} f_{k+1} & f_{k+1} + f_k \\ f_{k+3} & f_{k+2} + f_{k+1} \end{pmatrix}$$

$$= \det \left[\begin{pmatrix} f_k & f_{k+1} \\ f_{k+2} & f_{k+3} \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \right] = (-1)^{k+1} \cdot (-1) = (-1)^{k+2} \quad \square$$

$$\det(AB) = \det(A) \cdot \det(B)$$

what about 3×3 matrices?

an example:

$$\det \begin{pmatrix} 1 & 1 & 2 \\ 3 & 5 & 8 \\ 13 & 21 & 34 \end{pmatrix} = 1 \cdot \begin{vmatrix} 5 & 8 \\ 21 & 34 \end{vmatrix} - 1 \cdot \begin{vmatrix} 3 & 8 \\ 13 & 34 \end{vmatrix} + 2 \cdot \begin{vmatrix} 3 & 5 \\ 13 & 21 \end{vmatrix} \\ = 2 - (-2) + 2 \cdot (-2) = 0$$

$$\det \begin{pmatrix} f_n & f_{n+1} & f_{n+2} \\ f_{n+3} & f_{n+4} & f_{n+5} \\ f_{n+6} & f_{n+7} & f_{n+8} \end{pmatrix} = \det \begin{pmatrix} f_n & f_{n+1} & f_{n+1}+f_{n+2} \\ f_{n+3} & f_{n+4} & f_{n+3}+f_{n+4} \\ f_{n+6} & f_{n+7} & f_{n+6}+f_{n+7} \end{pmatrix}$$

$$= \det \left[\begin{pmatrix} f_n & f_{n+1} & f_n \\ f_{n+3} & f_{n+4} & f_{n+3} \\ f_{n+6} & f_{n+7} & f_{n+6} \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \right] = 0$$

And this holds for higher dimensions too, because with more than 2 rows you have at least 2 linear dependent rows. And this implies $\det = 0$. (quadratic matrices)

proof:

Suppose the rows of A are linearly dependent. Then at least two rows can be written as a linear combination. So if A has rows $r_1, \dots, r_n$, then $\exists c_1, \dots, c_n$, not all 0, such that

$$c_1 r_1 + \dots + c_n r_n = 0$$

$$\text{OE dA } r_n = c_1 r_1 + \dots + c_{n-1} r_{n-1}$$

$\det(A)$ is multilinear and additive. And, whenever a matrix has 2 identical rows/linear dependent rows, its $\det$ is zero due to the antisymmetric of $\det$; $\det A = 0$.

cause if you swap two rows then the $\det$ of A must change sign. But if you swap 2 identical rows, the $\det$ stays the same $\rightarrow$ hence it must be 0.

On the other hand: if $\det A = 0$, then A is not invertible. So there is no unique solution to the equation $Ax = 0$.

$\Rightarrow \exists x \neq 0$ with $Ax = 0 \Rightarrow$ the rows of A must be linearly dependent.